

PSG POLYTECHNIC COLLEGE INFORMATION TECHNOLOGY VIRTUALIZATION

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01. INTRODUCTION

Introduction about Virtualization

Introduction

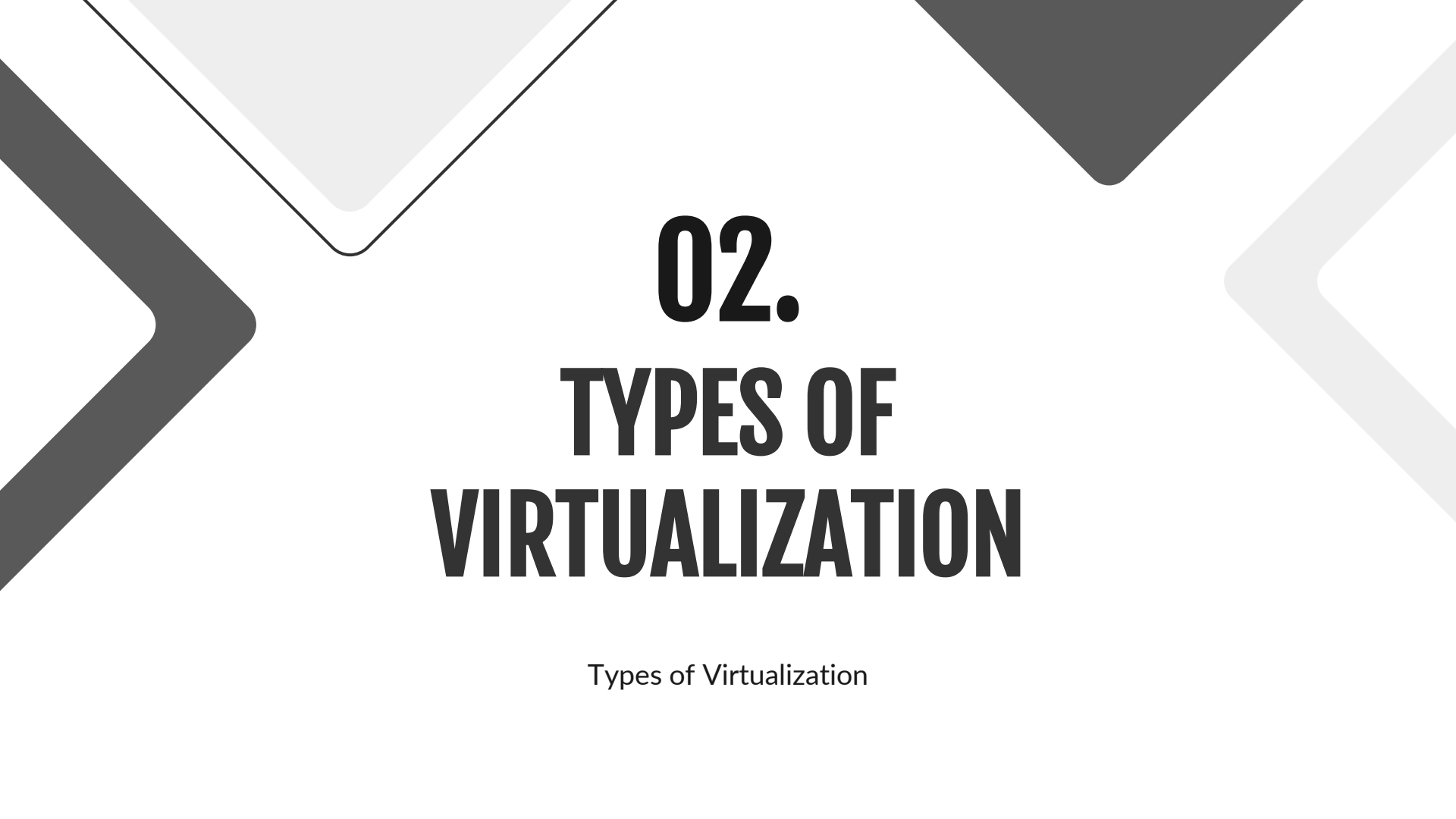
Virtualization is the technology that allows the creation of virtual environments or machines within a physical infrastructure, providing benefits such as resource optimization, flexibility, security, and efficient management of computing resources. It enables the abstraction and isolation of software from the underlying hardware, revolutionizing the way IT infrastructure is utilized and managed



DEFINITION

Virtualization is the creation of virtual instances that mimic physical computing resources, enabling efficient utilization, flexibility, and simplified management of IT infrastructure.





02. **TYPES OF** **VIRTUALIZATION**

Types of Virtualization

Types



Full-virtualization

Microsoft and Parallels systems are examples of full virtualization.



Para-virtualization

VMware and Xen are some examples of paravirtualization.



OS-level virtualization

Linux-VServer, Parallels Virtuozzo Containers some examples of os-level virtualization

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Full-Virtualization

Full Virtualization was introduced by IBM in the year 1966. It is the first software solution for server virtualization and uses binary translation and direct approach techniques. In full virtualization, the guest OS is completely isolated by the virtual machine from the virtualization layer and hardware. Microsoft and Parallels systems are examples of full virtualization.





Para-Virtualization

Paravirtualization is the category of CPU virtualization that uses hypercalls for operations to handle instructions at compile time. In paravirtualization, guest OS is not completely isolated but it is partially isolated by the virtual machine from the virtualization layer and hardware. VMware and Xen are some examples of paravirtualization.





OS level-Virtualization

OS virtualization is the use of software to allow a piece of hardware to run multiple operating system images at the same time. The technology got its start on mainframes decades ago, allowing administrators to avoid wasting expensive processing power.



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03. BENEFITS

Benefits of Virtualization

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BENEFITS

BENEFITS

Resource Optimization

Virtualization maximizes resource utilization by running multiple virtual machines on a single physical server, leading to cost savings and increased efficiency.

Cost Reduction

Virtualization reduces costs by consolidating servers and minimizing hardware and operational expenses. It streamlines management, reducing administrative overhead and labor costs.

CC ENABLEMENT

Virtualization is crucial for cloud computing, enabling scalable resource delivery and efficient sharing of infrastructure.



04.

CLOUD COMPUTING

Cloud computing in Virtualization

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CC IN VIRTUALIZATION

Multi-Tenancy and Resource Sharing

Virtualization allows for the efficient sharing of physical computing resources among multiple tenants in a cloud environment

Scalability and Elasticity

Virtualization enables scalable and elastic cloud computing through rapid provisioning.

Agility and Flexibility

Virtualization enhances cloud platform agility through seamless VM migration and efficient resource allocation, enabling improved availability and responsiveness to business needs.



05. TOOLS

Management and tools of Virtualization

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Tools and management

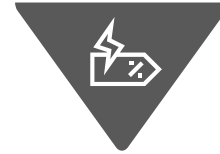
Virtualization management tools are designed to facilitate the administration, monitoring, and control of virtualized environments. These tools provide a range of functionalities to streamline virtual infrastructure management, optimize resource utilization, ensure performance and availability, and simplify day-to-day administrative tasks. Here are some commonly used virtualizations management tools:

Tools in Virtualization



SCVMM

SCVMM is a Microsoft Hyper-V virtualization platform management tool. It offers centralized management of virtual machines, virtual networking, and storage.



Proxmox VE

Proxmox VE is an open-source virtualization management platform that combines KVM and LXC containerization. It provides a web-based management

VMware vSphere

VMware vSphere is a powerful virtualization platform for data centers, offering robust tools for managing and deploying virtual machines.



Citrix XenCenter

XenCenter is a management console for Citrix XenServer, an open-source virtualization platform. It provides a user-friendly interface to manage virtual machines, storage, and networking resources.



06. FUTURE TRENDS

Future of Virtualization

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Containerization

Containerization technology, led by platforms like Docker and Kubernetes, is gaining popularity as an alternative to traditional virtualization. Containers offer lightweight, isolated environments for running applications, allowing for greater efficiency, faster startup times, and improved scalability.





Serverless Computing

Serverless computing, also known as Function as a Service (FaaS), abstracts away the need to manage infrastructure resources and focuses on running individual functions or tasks. Virtualization plays a crucial role in serverless platforms by dynamically provisioning and managing the underlying resources needed to execute functions.



AI and Machine Learning

Artificial Intelligence (AI) and Machine Learning (ML) techniques are being leveraged to optimize virtualization and resource management. AI-powered algorithms can analyze workload patterns, resource usage, and performance data to dynamically allocate and optimize resources based on demand, leading to improved efficiency, cost savings, and better performance in virtualized environments.





My Project

Drag and Drop & Virtual Steering

BY
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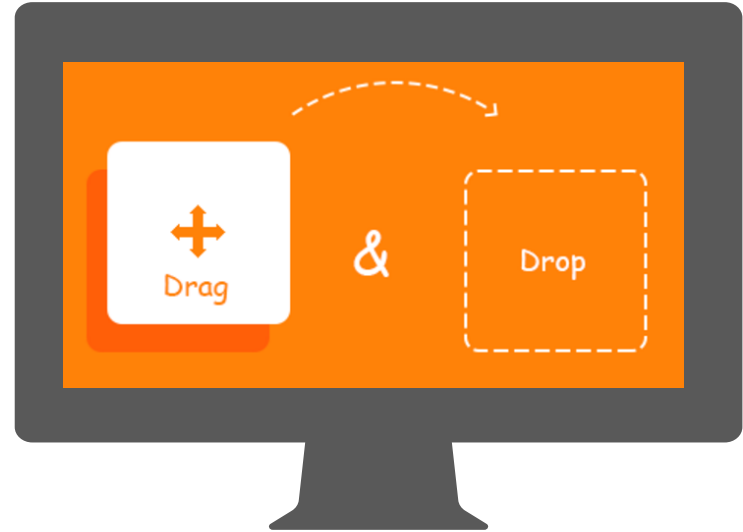
Project 01

Drag & Drop

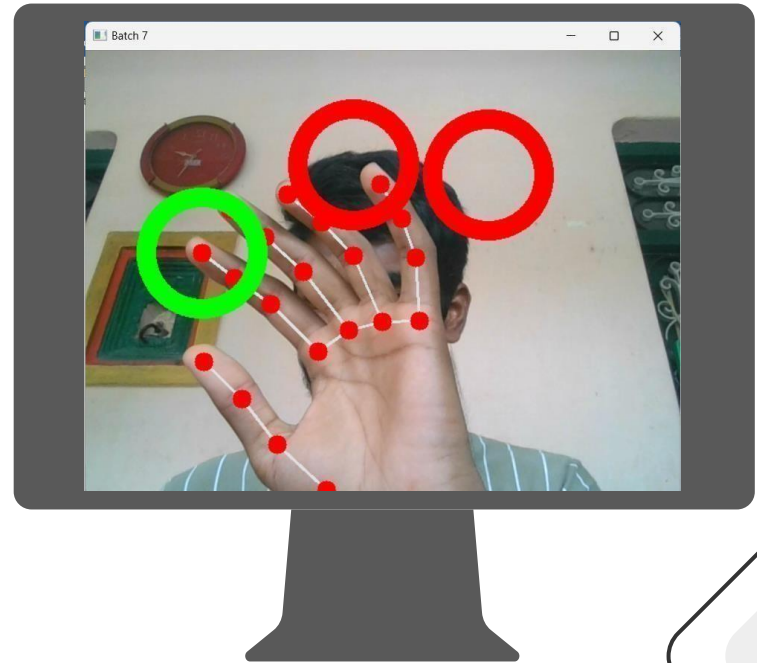
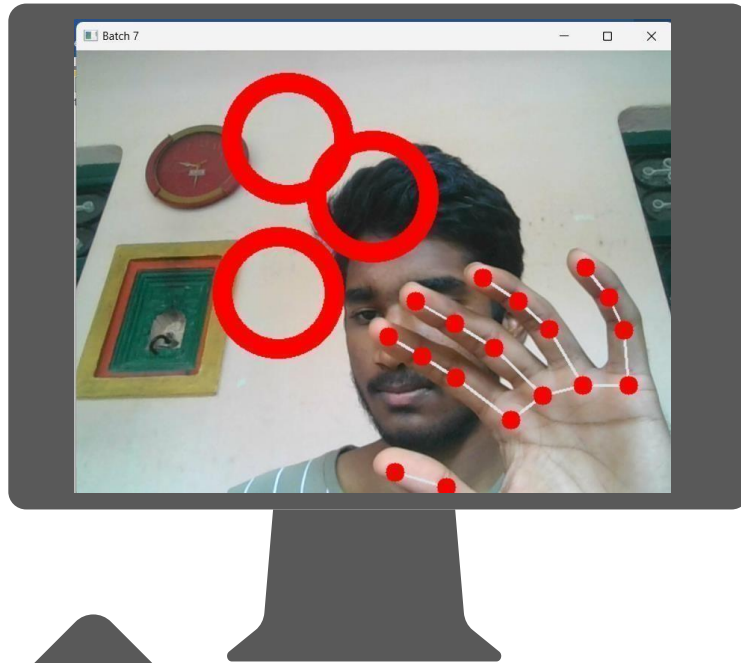
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Python 3.10.11

Drop & Drop

My Project on virtualization is a Python program that makes an object drag and drop through hand gestures by handling the camera portion.



OUTPUTS



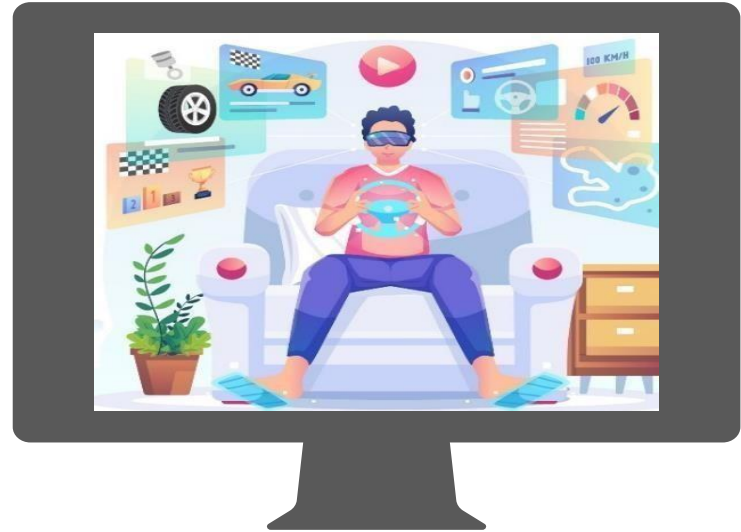
Project 02

VIRTUAL STEERING

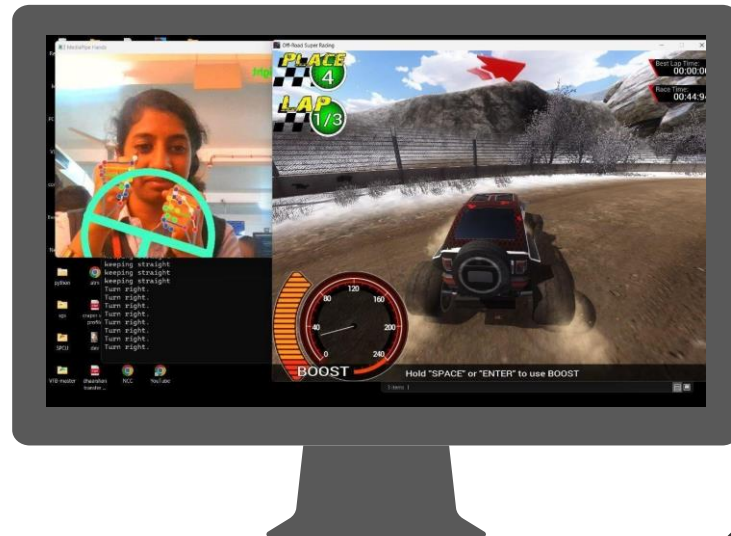
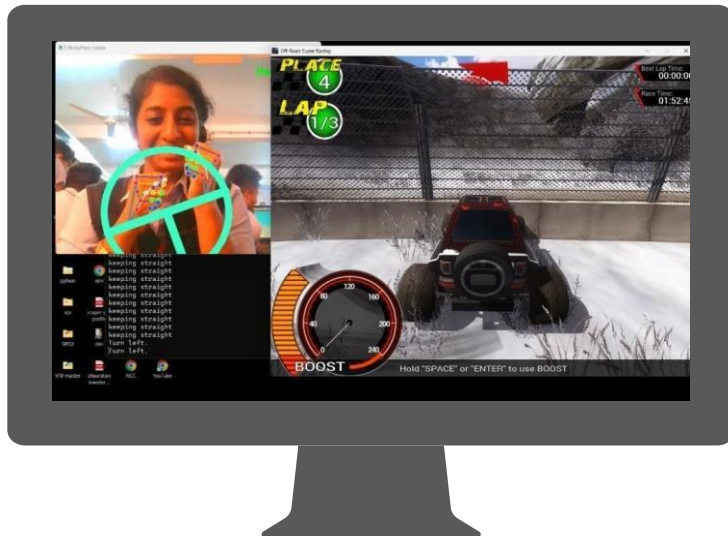
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Virtual Steering

Virtual steering refers to the simulation of steering a vehicle in a digital or virtual environment, typically using input devices and software to replicate the experience. It is commonly used in video games, virtual reality simulations, and driving training programs to provide users with an immersive and realistic driving experience



OUTPUTS



OUTPUTS



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THANK YOU